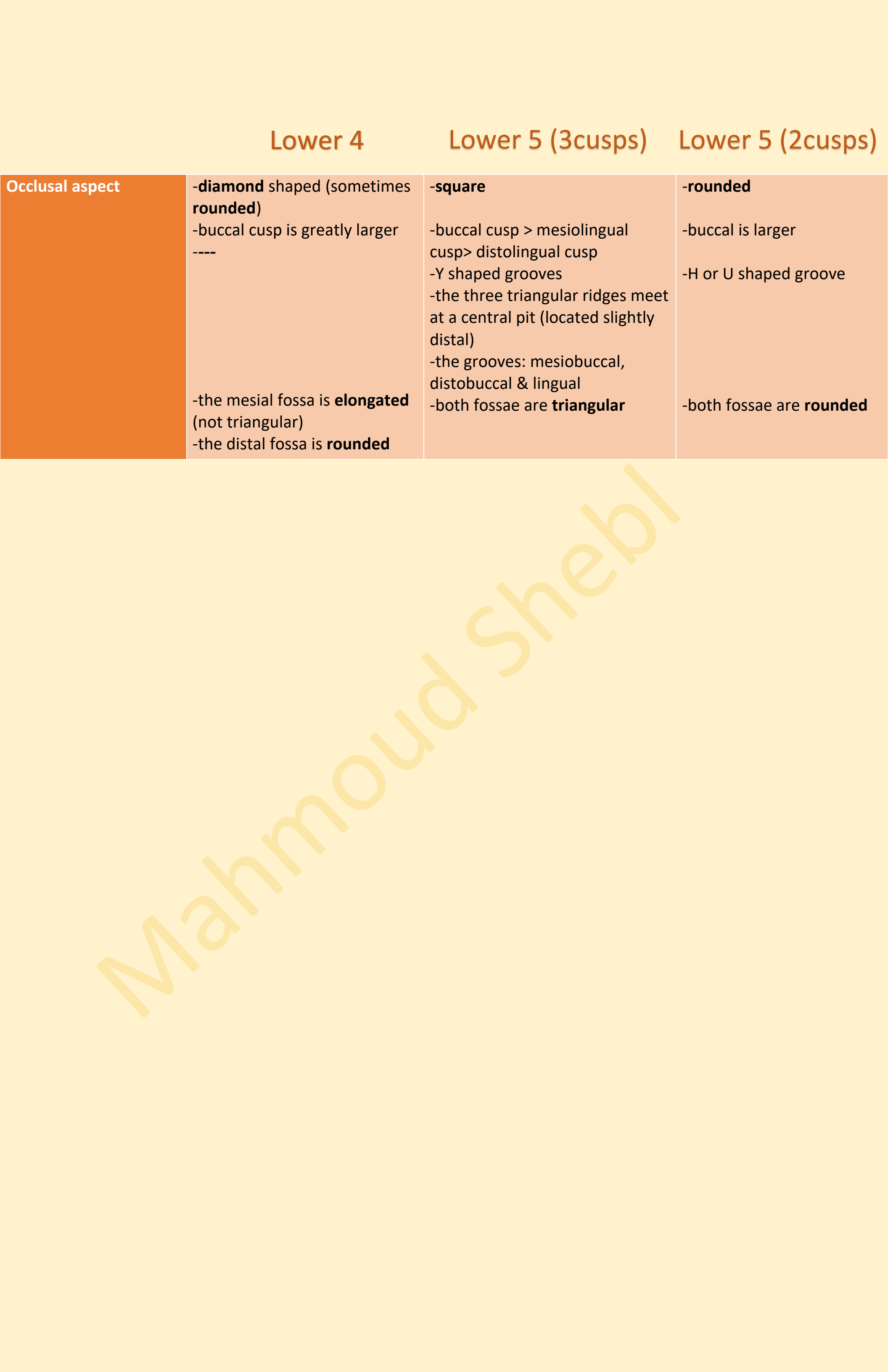


# Mandibular premolars

|                                 | mandibular first premolar                                                                                                                                                                                                                                                                                                                                                | mandibular second premolar                                                                                                                                                                                                               |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mesial outline                  | -Slightly <b>concave</b><br>Mesial slope: slightly <b>concave</b>                                                                                                                                                                                                                                                                                                        | <b>The same</b>                                                                                                                                                                                                                          |
| Mesial contact area             | Slightly occlusal to the <b>middle</b> of the crown<br>(considering the cusp tip shifted mesially).<br><b>OR</b><br>Slightly occlusal to the <b>occlusal/middle junction</b> (considering the cusp tip at the midline)                                                                                                                                                   | Appear more cervical (due to the cusp slopes meeting in a sharper angle)<br><br><b>centered buccolingually</b>                                                                                                                           |
| Distal outline                  | -slightly <b>concave</b><br>Distal slope: slightly <b>concave</b>                                                                                                                                                                                                                                                                                                        | <b>The same</b>                                                                                                                                                                                                                          |
| Distal contact area             | Nearly the <b>same level</b><br>(considering the cusp tip shifted mesially)<br><b>OR</b><br>At the <b>occlusal/middle junction</b><br>(considering the cusp tip at the midline)<br><br><b>Broader, situated buccally</b>                                                                                                                                                 | the same as the mesial one or more cervically<br><br><b>Broader, situated buccally</b>                                                                                                                                                   |
| buccal outline (proximal view)  | Convex, with the maximum convexity...<br>The buccal cusp tip is at the <b>central line of the root</b>                                                                                                                                                                                                                                                                   | <b>The same</b><br>With the buccal cusp tip lies <b>buccally</b> to the central line ( <b>midline</b> ) of the root                                                                                                                      |
| Lingual outline (proximal view) | Convex with the maximum convexity on the middle third. The cusp tip is in line with the <b>lingual border</b> of the root                                                                                                                                                                                                                                                | The maximum convexity on the <b>occlusal third</b> . The cusp tip is in line with the <b>lingual border</b> of the root                                                                                                                  |
| The root                        | • <i>buccal &amp; lingual views</i> :<br>-conical with <b>pointed apex</b> and often curves distally<br>• <i>proximal views</i> :<br>-the same outline as canines<br>- <b>deep mesial developmental groove</b><br>- <b>distal developmental depression</b><br>- <b>rarely distal developmental groove</b><br>-the root surface is more convex distally                   | • <i>buccal &amp; lingual views</i> :<br>Conical with <b>Blunt apex</b><br>Less likely distal inclination<br>Characteristic with a <b>deep buccal DG</b><br>• <i>proximal views</i> :<br>The same outline but no grooves or depressions  |
| Pulp cavity (chamber & canal)   | • <b>mesiodistal section</b> :<br>- Has narrow pulp chamber & root canal<br>• <b>labiolingual section</b> :<br>-shows wider pulp chamber (buccolingually) with two pulp horns<br>-the root canal is wide & becomes constricted at the apical third<br>• <b>midroot cross section</b> :<br>-circular & centered                                                           | <b>The same</b><br>With three pulp horns rather than 2 in 3 cusp type                                                                                                                                                                    |
| features                        | -more prominent buccal ridge<br><br>- <b>mesial &amp; distal buccal DD</b> exist but less observed than the maxillary premolars’<br>-the lingual cusp is <b>greatly shorter</b><br>-mesiolingual groove<br><br>-mesial marginal ridge <b>slopes downwards</b> & thus more of the occlusal aspect is visible from the mesial view.<br>The distal one is <b>horizontal</b> | -less prominent<br><br><br><br><b>the same</b><br>-the lingual cusp(s) is <b>slightly shorter</b><br>-lingual groove between the two lingual cusps<br>-distal marginal ridge is lower than the mesial one but both are <b>horizontal</b> |



|                 | Lower 4                                                                                                                                                                                                                                                          | Lower 5 (3cusps)                                                                                                                                                                                                                                                                                                                                        | Lower 5 (2cusps)                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Occlusal aspect | <ul style="list-style-type: none"><li>-<b>diamond</b> shaped (sometimes <b>rounded</b>)</li><li>-buccal cusp is greatly larger</li><li>----</li><li>-the mesial fossa is <b>elongated</b> (not triangular)</li><li>-the distal fossa is <b>rounded</b></li></ul> | <ul style="list-style-type: none"><li>-<b>square</b></li><li>-buccal cusp &gt; mesiolingual cusp&gt; distolingual cusp</li><li>-Y shaped grooves</li><li>-the three triangular ridges meet at a central pit (located slightly distal)</li><li>-the grooves: mesiobuccal, distobuccal &amp; lingual</li><li>-both fossae are <b>triangular</b></li></ul> | <ul style="list-style-type: none"><li>-<b>rounded</b></li><li>-buccal is larger</li><li>-H or U shaped groove</li><li>-both fossae are <b>rounded</b></li></ul> |